24.2: Speciation can take place with or without geographic separation

Having discussed what constitutes a unique species, let's return to the process by which such species arise from existing species. We'll describe this process by focusing on the geographic setting in which gene flow is interrupted between populations of the existing species—in allopatric speciation the populations are geographically isolated, while in sympatric speciation they are not **(Figure 24.5)**.

Figure 24.5

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The geography of speciation.
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Gold fish in a pond with a barrier partially submerged. If the pools are divided, allopatric speciation can occur. If the pools remain unchanged sympatric speciation can occur.

Gold fish are in a pond with a barrier in the middle that is partially submerged. If the water level drops, the pools are divided and allopatric speciation can occur which is shown as two blue fish in the left pool and five gold fish in the right pool. If the water level remains unchanged the fish are not divided and sympatric speciation can occur which is shown as two blue fish among eight gold fish.

- a. Allopatric speciation: A population forms a new species while geographically isolated from its parent population.
- b. Sympatric speciation: A subset of a population forms a new species without geographic separation.

Allopatric ("Other Country") Speciation

In **allopatric speciation** (from the Greek *allos*, other, and *patra*, homeland), gene flow is interrupted when a population is divided into geographically isolated subpopulations. For example, the water level in a lake may subside, resulting in two or more smaller lakes that are now home to separated populations (see Figure 24.5a). Or a river may change course and divide a population of animals that cannot cross it. Allopatric speciation can also occur without geologic change, such as when individuals colonize a remote area and their descendants become geographically isolated from the parent population. The flightless cormorant shown in Figure 24.1 probably originated in this way from an ancestral flying species that reached the Galápagos Islands.

The Process of Allopatric Speciation

How formidable must a geographic barrier be to promote allopatric speciation? The answer depends on the ability of the organisms to move about. Birds, mountain lions, and coyotes can cross rivers and canyons—as can windblown pollen and the seeds of some flowering plants. In contrast, small rodents may find a river or canyon a formidable barrier.

Once geographic isolation has occurred, the separated gene pools may diverge. Different mutations arise, and natural selection and genetic drift may alter allele frequencies in different ways in the separated populations. Reproductive isolation may then evolve as a by-product of the genetic divergence that results from selection or drift.

Figure 24.6 describes an example. On Andros Island, in the Bahamas, populations of the mosquitofish *Gambusia hubbsi* colonized a series of ponds that later became isolated from one another. Genetic analyses indicate that little or no gene flow currently occurs between the ponds. The environments of these ponds are very similar except that some contain predatory fishes, while others do not. In ponds with predatory fishes, selection has favored the evolution of a mosquitofish body shape that enables rapid bursts of speed (see Figure 24.6a). In ponds without predatory fishes, selection has favored a different body shape, one that improves the ability to swim for long periods of time (see Figure 24.6b).

Figure 24.6

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Evolution in mosquitofish populations.
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Evolution in mosquitofish populations.

Different body shapes have evolved in mosquitofish populations from ponds with and without predators. These differences affect how quickly the fishes can accelerate to escape and their survival rate when exposed to predators.

- Part a. Differences in body shape: High predation can lead to differences in body shape that enable quick getaways, for example in mosquitofish. In ponds with predatory fishes, the head of the mosquitofish is streamlined and the tail is powerful, enabling rapid bursts of speed. Under low predation. In ponds without predatory fishes, mosquitofish have a different body shape that favors long, steady swimming.
- Part b. Differences in escape acceleration and survival: mosquitofish in ponds containing predators show faster escape acceleration and higher survival rates than mosquitofish in ponds without predators.

Two bar graphs. Y-axis of graph on left is labeled escape acceleration (meters per s squared) and is marked from 50 to 110 in increments of 10. Mosquitofish from source ponds with predators present show an escape acceleration of almost 100 meters per s squared (orange bar). Mosquitofish from ponds with no predators show a much lower escape acceleration of approximately 70 meters per s squared (green bar). Y-axis of graph on right is labeled survival rate with predators and is marked from zero to 1.0 in increments of 0.2. Mosquitofish from ponds with predators present have a higher survival rate than those from ponds with no predators: 0.9 (orange bar) compared to 0.4 (green bar).

How have these different selective pressures affected the evolution of reproductive barriers? Researchers studied this question by bringing together mosquitofish from the two types of ponds. They found that female mosquitofish prefer to mate with males whose body shape is similar to their own. This preference establishes a behavioral barrier to reproduction between mosquitofish from ponds with predators and those from ponds without predators. Thus, as a by-product of selection for avoiding predators, reproductive barriers have formed in these allopatric populations.

Evidence of Allopatric Speciation

Many studies provide evidence that speciation can occur in allopatric populations. For example, laboratory studies show that reproductive barriers can develop when populations are isolated experimentally and subjected to different environmental conditions **(Figure 24.7)**.

Figure 24.7

Inquiry: Can divergence of allopatric populations lead to reproductive isolation?

Experiment

A researcher divided a laboratory population of the fruit fly *Drosophila pseudoobscura*, raising some flies on a starch medium and others on a maltose medium. After one year (about 40 generations), natural selection resulted in divergent evolution: Populations raised on starch digested starch more efficiently, while those raised on maltose digested maltose more efficiently. The researcher then put flies from the same or different populations in mating cages and measured mating frequencies. All flies used in the mating preference tests were reared for one generation on a standard cornmeal medium.

Can divergence of allopatric populations lead to reproductive isolation?

Experiment. A researcher divided a laboratory population of the fruit fly Drosophila pseudoobscura, raising some flies on a starch medium and others on a maltose medium. After one year (about 40 generations), natural selection resulted in divergent evolution: Populations raised on starch digested starch more efficiently, while those raised on maltose digested maltose more efficiently. The researcher then put flies from the same or different populations in mating cages and measured mating frequencies. All flies used in the mating preference tests were reared for one generation on a standard cornmeal medium.

Results. Mating patterns among populations of flies raised on different media are shown below. When flies from starch populations were mixed with flies from maltose populations, the flies tended to mate with like partners (see the first table below). But in a control group consisting of flies from two different populations adapted to starch, the flies were about as likely to mate with flies of the other population as with flies of their own population (see the second table below); similar results were obtained for control groups adapted to maltose.

Results

Mating patterns among populations of flies raised on different media are shown below. When flies from “starch populations” were mixed with flies from “maltose populations,” the flies tended to mate with like partners (see the lefthand table). But in a control group consisting of flies from two different populations adapted to starch, the flies were about as likely to mate with flies of the other population as with flies of their own population (see the righthand table); similar results were obtained for control groups adapted to maltose.

Results of an experiment on Can divergence of allopatric populations lead to reproductive isolation?

	Female
Starch	Maltose
Male	Starch
Maltose	8
Number of matings in experimental group	

	Female
Starch population 1	Starch population 2
Male	Starch population 1

Starch population 2	12
Number of matings in control group	

Conclusion

In the experimental group, the strong preference of “starch flies” and “maltose flies” to mate with like-adapted flies indicates that a reproductive barrier was forming between these fly populations. Although this barrier was not absolute (some mating between starch flies and maltose flies did occur), after 40 generations reproductive isolation appeared to be increasing. This barrier may have been caused by differences in courtship behavior that arose as an incidental by-product of differing selective pressures as these allopatric populations adapted to different sources of food.

Data from D. M. B. Dodd, Reproductive isolation as a consequence of adaptive divergence in *Drosophila pseudoobscura, Evolution* 43:1308–1311 (1989).

WHAT IF?_Why were all flies used in the mating preference tests reared on a standard medium (rather than on starch or maltose)?_

For suggested answer, see*Appendix A.*

Field studies indicate that allopatric speciation also can occur in nature. Consider the 30 species of snapping shrimp in the genus *Alpheus* that live off the Isthmus of Panama, the land bridge that connects South and North America **(Figure 24.8)**. Fifteen of these species live on the Atlantic side of the isthmus, while the other 15 live on the Pacific side. Before the isthmus formed, gene flow could occur between the Atlantic and Pacific populations of snapping shrimp. Did the species on different sides of the isthmus originate by allopatric speciation? Morphological and genetic data group these shrimp into 15 pairs of *sister species*, pairs whose member species are each other’s closest relative. In each of these 15 pairs, one of the sister species lives on the Atlantic side of the isthmus, while the other lives on the Pacific side. This fact strongly suggests that the two species arose as a consequence of geographic separation. Furthermore, genetic analyses indicate that the *Alpheus* species originated from 9 to 3 million years ago, with the sister species that live in the deepest water diverging first. These divergence times are consistent with geologic evidence that the isthmus formed gradually, starting 10 million years ago, and closing completely about 3 million years ago.

Figure 24.8

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Allopatric speciation in snapping shrimp (*Alpheus*).
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A map of the Isthmus of Panama is shown with A formosus and A nuttingi shrimp in the Atlantic Ocean, and A panamensis (sister A formosus) and A millsae (sister A nuttingi) in the Pacific Ocean.

The shrimp pictured are just 2 of the 15 pairs of sister species that arose as populations were divided by the formation of the Isthmus of Panama. The color-coded type indicates the sister species.

The importance of allopatric speciation is also suggested by the fact that regions that are isolated or highly subdivided by barriers typically have more species than do otherwise similar regions that lack such features. For example, many unique plants and animals are found on the geographically isolated Hawaiian Islands (we'll return to the origin of Hawaiian species in Concept 25.4). Field studies also show that reproductive isolation between two populations generally increases as the geographic distance between them increases, a finding consistent with allopatric speciation. In the **Scientific Skills Exercise**, you can analyze data from one such study that examined reproductive isolation in geographically separated salamander populations.

Scientific Skills Exercise

Identifying Independent and Dependent Variables, Making a Scatter Plot, and Interpreting Data

AP Science Practices 2, 3, 4, 5

Does Distance Between Salamander Populations Increase Their Reproductive Isolation? Allopatric speciation begins when populations become geographically isolated, reducing mating between individuals in different populations and thus reducing gene flow. It is logical that as distance between populations increases, so will their degree of reproductive isolation. To test this hypothesis, researchers studied populations of the dusky salamander (*Desmognathus ochrophaeus*) living on different mountain ranges in the southern Appalachians.

A salamander.

How the Experiment Was Done The researchers tested the reproductive isolation of pairs of salamander populations by leaving one male and one female together and later checking the females for the presence of sperm. Four mating combinations were tested for each pair of populations (A and B)—two *within* the same population (female A with male A and female B with male B) and two *between* populations (female A with male B and female B with male A).

Data from the Experiment The researchers used an index of reproductive isolation that ranged from a value of 0 (no isolation) to a value of 2 (full isolation). The proportion of successful matings for each mating combination was measured, with 100% success=1 and no success=0. The reproductive isolation value for two populations is the sum of the proportion of successful matings of each type within populations (AA+BB) minus the sum of the proportion of successful matings of each type between populations (AB+BA). The table provides distance and reproductive isolation data for 27 pairs of dusky salamander populations.

A table lists geographic distance and reproductive isolation values.

Geographic distance, kilometers	15	32	40	47	42	62	63	81	86	107	107
Reproductive isolation value	0.32	0.54	0.5	0.5	0.82	0.37	0.67	0.53	1.15	0.73	0.82
Distance (continued)	137	150	165	189	219	239	247	53	55	62	105
Isolation (continued)	0.5	0.57	0.91	0.93	1.5	1.22	0.82	0.99	0.21	0.56	0.41

Data from S. G. Tilley, A. Verrell, and S. J. Arnold, Correspondence between sexual isolation and allozyme differentiation: a test in the salamander *Desmognathus ochrophaeus, Proceedings of the National Academy of Sciences USA* 87:2715–2719 (1990).

INTERPRET THE DATA

- State the researchers' hypothesis, and identify the independent and dependent variables in this study. Explain why the researchers used four mating combinations for each pair of populations.
- Calculate the value of the reproductive isolation index
- if *all* of the matings within a population were successful, but *none* of the matings between populations were successful;
- if salamanders are equally successful in mating with members of their own population and members of another population.
- Make a scatter plot to help you visualize any patterns that might indicate a relationship between the variables. Plot the independent variable on the x-axis and the dependent variable on the y-axis. (For additional information about graphs, see the Scientific Skills Review in Appendix D.)

- Interpret your graph by
- explaining in words any pattern indicating a possible relationship between the variables and
- hypothesizing the possible cause of such a relationship.

Instructors: A version of this Scientific Skills Exercise can be assigned in **Mastering Biology**.

Note that while geographic isolation prevents interbreeding between members of allopatric populations, physical separation is not a biological barrier to reproduction. Biological reproductive barriers such as those described in Figure 24.3 are intrinsic to the organisms themselves. Hence, it is biological barriers that can prevent interbreeding when members of different populations come into contact with one another.

Sympatric ("Same Country") Speciation

In **sympatric speciation** (from the Greek *syn*, together), speciation occurs in populations that live in the same geographic area (see Figure 24.5b). How can reproductive barriers form between sympatric populations while their members remain in contact with each other? Although such contact (and the ongoing gene flow that results) makes sympatric speciation less common than allopatric speciation, sympatric speciation can occur if gene flow is reduced by such factors as polyploidy, sexual selection, and habitat differentiation. (Note that sexual selection and habitat differentiation also can promote allopatric speciation.)

Polyploidy

A species may originate from an accident during cell division that results in extra sets of chromosomes, a condition called **polyploidy**. Polyploid speciation occasionally occurs in animals; for example, the gray tree frog *Hyla versicolor* (see Figure 23.16) is thought to have originated in this way. However, polyploidy is far more common in plants. In fact, botanists estimate that more than 80% of the plant species alive today are descended from ancestors that formed by polyploid speciation.

Two distinct forms of polyploidy have been observed in plant (and a few animal) populations. An **autopolyploid** (from the Greek *autos*, self) is an individual that has more than two chromosome sets that are all derived from a single species. In plants, for example, a failure of cell division could double a cell's chromosome number from the original number (2n) to a tetraploid number (4n) **(Figure 24.9)**.

Figure 24.9

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Sympatric speciation by autopolyploidy.
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Autopolyploid speciation. Cell division error leads to new species (4 n).

Cell is 2 n = 6 (3 sets of chromosomes). After cell division error, tetraploid cell results (4 n): there are four of each chromosome instead of two, for 12 total chromosomes. Tetraploid cell undergoes meiosis and the gametes are 2 n = 6. After fertilization, a new 4 n species results with 12 chromosomes.

A tetraploid can produce fertile tetraploid offspring by self-pollinating or by mating with other tetraploids. In addition, the tetraploids are reproductively isolated from 2n plants of the original population because the triploid (3n) offspring of such unions have reduced fertility. Thus, in just one generation, autopolyploidy can generate reproductive isolation without any geographic separation.

A second form of polyploidy can occur when two different species interbreed and produce hybrid offspring. Most such hybrids are sterile because the set of chromosomes from one species cannot pair during meiosis with the set of chromosomes from the other species. However, an infertile hybrid may be able to propagate itself asexually (as many plants can do). In subsequent generations, various mechanisms can change a sterile hybrid into a fertile polyploid called an **allopolyploid(Figure 24.10)**. The allopolyploids are fertile when mating with each other but cannot interbreed with either parent species; thus, they represent a new biological species. The new species has a diploid chromosome number equal to the sum of the diploid chromosome numbers of the two parent species.

Figure 24.10

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One mechanism for allopolyploid speciation in plants.
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Gametes from Species A and Species B form a sterile hybrid with 5 chromosomes. Mitotic or meiotic error doubles the chromosome number and a new viable, fertile species is formed.

Most hybrids are sterile because their chromosomes are not homologous and cannot pair during meiosis. However, such a hybrid may be able to reproduce asexually. This diagram traces one mechanism by which sterile hybrids can give rise to fertile allopolyploids that are members of a new species.

Species A (2 n = 6) shown with a normal gamete (n = 3) along with Species B (2 n = 4) and normal gamete (n = 2). A sterile hybrid is formed with 5 chromosomes. Mitotic or meiotic error doubles the chromosome number and a new species is formed. The new species is a viable, fertile hybrid (allopolyploid) with 2 n = 10.

Figure 24.10 Walkthrough

One mechanism for allopolyploid speciation in plants

Watch an explanation of this figure in the Figure 24.10 Walkthrough: One mechanism for allopolyploid speciation in plants.

image

Play

image

Play

Although it can be challenging to study speciation in the field, scientists have documented at least five new plant species that have originated by polyploid speciation since 1850. One of these examples involves the origin of a new species of goatsbeard plant (genus *Tragopogon*) in the Pacific Northwest. *Tragopogon* first arrived in the region when humans introduced three European species in the early 1900s: *T. pratensis, T. dubius*, and *T. porrifolius*. These three species are now common in the Pacific Northwest and many other parts of North America. In 1950, a new *Tragopogon* species was discovered near the Idaho-Washington border, a region where all three European species also were found. Genetic analyses revealed that this new species, *Tragopogon miscellus*, is an allopolyploid that originated from a hybrid of two of the European species **(Figure 24.11)**. Although the *T. miscellus* population grows mainly by reproduction of its own members, hybridization between the parent species (followed by a mitotic or meiotic error that doubles the chromosome number) continues to add new members to the *T. miscellus* population. Later, scientists discovered another new *Tragopogon* species, *T. mirus*, which also arose by polyploid speciation.

Figure 24.11

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Allopolyploid speciation in *Tragopogon*.
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Parent species T dubius (12) and T pratensis (12) form hybrid species T miscellus (24). Parent T dubius (12) and T porrifolius (12) form hybrid T mirus (24).

The gray boxes indicate the parent species of the two new polyploid species. The diploid chromosome number of each species is shown in parentheses.

Parent species T dubius (12) and T pratensis (12), both with yellow flowers, form hybrid species T miscellus (24) which also has yellow flowers. Parent T dubius (12) with yellow flowers and T porrifolius (12) with purple flowers form hybrid T mirus (24) with pink flowers.

VISUAL SKILLS *Based on this diagram, identify the two parent species of each polyploid species.*

*For suggested answer, see*Appendix A*.*

Many important agricultural crops—such as oats, cotton, potatoes, tobacco, and wheat—are polyploids. For example, the wheat used for bread, *Triticum aestivum*, is an allohexaploid (six sets of chromosomes, two

sets from each of three different species). The events that eventually led to modern wheat probably began about 8,000 years ago in the Middle East when an early cultivated wheat species hybridized with a wild grass. Today, plant geneticists generate new polyploids in the laboratory by using chemicals that induce meiotic and mitotic errors. By harnessing the evolutionary process, researchers can produce new polyploid species with desired qualities, such as a polyploid that combines the high yield of wheat with the hardiness of rye.

Animation

Speciation by Changes in Ploidy

image

Play

image

Sexual Selection

There is evidence that sympatric speciation can also be driven by sexual selection. Clues to how this can occur have been found in cichlid fish from one of Earth's hot spots of animal speciation, East Africa's Lake Victoria. This lake was once home to as many as 600 species of cichlids. Genetic data indicate that these species originated within the last 100,000 years from a small number of colonizing species that arrived from other lakes and rivers. How did so many species—more than double the number of freshwater fish species known in all of Europe—originate within a single lake?

One hypothesis is that subgroups of the original cichlid populations adapted to different food sources and the resulting genetic divergence contributed to speciation in Lake Victoria. But sexual selection, in which (typically) females select males based on their appearance (see Concept 23.4), may also have been a factor. Researchers have studied two closely related sympatric species of cichlids that differ mainly in the coloration of breeding males: Breeding *Pundamilia pundamilia* males have a blue-tinged back, whereas breeding *Pundamilia nyererei* males have a red-tinged back **(Figure 24.12)**. Their results suggest that mate choice based on male breeding coloration can act as a reproductive barrier that keeps the gene pools of these two species separate.

Figure 24.12

Inquiry: Does sexual selection in cichlids result in reproductive isolation?

Experiment

Researchers placed males and females of *Pundamilia pundamilia* and *P. nyererei* together in two aquarium tanks, one with natural light and one with a monochromatic orange lamp. Under normal light, the two species are noticeably different in male breeding coloration; under monochromatic orange light, the two species are very similar in color. The researchers then observed the mate choices of the females in each tank.

Does sexual selection in cichlids result in reproductive isolation?

Experiment. Researchers placed males and females of Pundamilia pundamilia and P nyererei together in two aquarium tanks, one with natural light and one with a monochromatic orange lamp. Under normal light, the two species are noticeably different in male breeding coloration; under monochromatic orange light, the two species are very similar in color. The researchers then observed the mate choices of the females in each tank. Results. Under normal light, females of each species strongly preferred males of their own species. But under orange light, females of each species responded indiscriminately to males of both species. The resulting hybrids were viable and fertile.

Conclusion. The researchers concluded that mate choice by females based on male breeding coloration can act as a reproductive barrier that keeps the gene pools of these two species separate. Since the species can still interbreed when this prezygotic barrier is breached in the laboratory, the genetic divergence between the species is likely to be small. This suggests that speciation in nature has occurred relatively recently.

Results

Under normal light, females of each species strongly preferred males of their own species. But under orange light, females of each species responded indiscriminately to males of both species. The resulting hybrids were viable and fertile.

Conclusion

The researchers concluded that mate choice by females based on male breeding coloration can act as a reproductive barrier that keeps the gene pools of these two species separate. Since the species can still interbreed when this prezygotic barrier is breached in the laboratory, the genetic divergence between the species is likely to be small. This suggests that speciation in nature has occurred relatively recently.

Data from O. Seehausen and J. J. M. van Alphen, The effect of male coloration on female mate choice in closely related Lake Victoria cichlids (*Haplochromis nyererei* complex), *Behavioral Ecology and Sociobiology* 42:1–8 (1998).

WHAT IF? *Suppose that female cichlids living in the murky waters of a polluted lake could not distinguish colors well. In such waters, how might the gene pools of these species change over time?*

*For suggested answer, see*Appendix A*.*

Habitat Differentiation

Sympatric speciation can also occur when a subpopulation exploits a habitat or resource not used by the parent population. Consider the North American apple maggot fly (*Rhagoletis pomonella*), a pest of apples. The fly's original habitat was the native hawthorn tree (see Figure 24.3a), but about 200 years ago, some populations colonized apple trees that had been introduced by European settlers. Apple maggot flies usually mate on or near their host plant. This results in a prezygotic barrier (habitat isolation) between populations that feed on apples and populations that feed on hawthorns. Furthermore, because apples mature more quickly than hawthorn fruit, natural selection has favored apple-feeding flies with rapid development. These apple-feeding populations now show temporal isolation from the hawthorn-feeding *R. pomonella*, providing a second prezygotic barrier to gene flow between the two populations. Researchers also have identified alleles that benefit the flies that use one host plant but harm the flies that use the other host plant. Natural selection operating on these alleles has provided a postzygotic barrier to reproduction, further limiting gene flow. Altogether, although the two populations are still classified as subspecies rather than separate species, sympatric speciation appears to be well under way.

Allopatric and Sympatric Speciation: *A Review*

Now let's recap the processes by which new species form. In allopatric speciation, a new species forms in geographic isolation from its parent population. Geographic isolation severely restricts gene flow. Intrinsic barriers to reproduction with members of the parent population may then arise as a by-product of genetic changes that occur within the isolated population. Many different processes can produce such genetic changes, including natural selection under different environmental conditions, genetic drift, and sexual selection. Once formed, reproductive barriers that arise in allopatric populations can prevent interbreeding with the parent population even if the populations come back into contact.

Sympatric speciation, in contrast, requires the emergence of a reproductive barrier that isolates a subset of a population from the remainder of the population in the same area. Though rarer than allopatric speciation, sympatric speciation can occur when gene flow to and from the isolated subpopulation is blocked. This can

occur as a result of polyploidy, a condition in which an organism has extra sets of chromosomes. Sympatric speciation also can result from sexual selection. Finally, sympatric speciation can occur when a subset of a population becomes reproductively isolated because of natural selection that results from a switch to a habitat or food source not used by the parent population.

Having reviewed the geographic context in which species originate, we'll next explore in more detail what can happen when new or partially formed species come into contact.

HHMI BioInteractive Video

Anole Lizards: An Example of Speciation

image

Play

image

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